Machine Learning with Python

Hajar Bougataya

December 2025

Phase 0 — Data preparation and exploratory analysis

Q0.1 — Target variable and class imbalance

The target variable `income` is binary:

- `<=50K`
- `>50K`

The observed class distribution is:

- `<=50K`: 75.9%
- `>50K`: 24.1%

This moderate imbalance motivates the use of metrics beyond accuracy, namely balanced accuracy, F1-score and ROC-AUC.

Q0.2 — Data cleaning and preprocessing

Missing values encoded as `"?"` were identified in `workclass`, `occupation` and `native_country`. They were replaced by explicit missing values and imputed with the category `Unknown`. Column names were standardized and categorical values were cleaned to ensure compatibility with scikit-learn pipelines.

Q0.3 — Exploratory data analysis

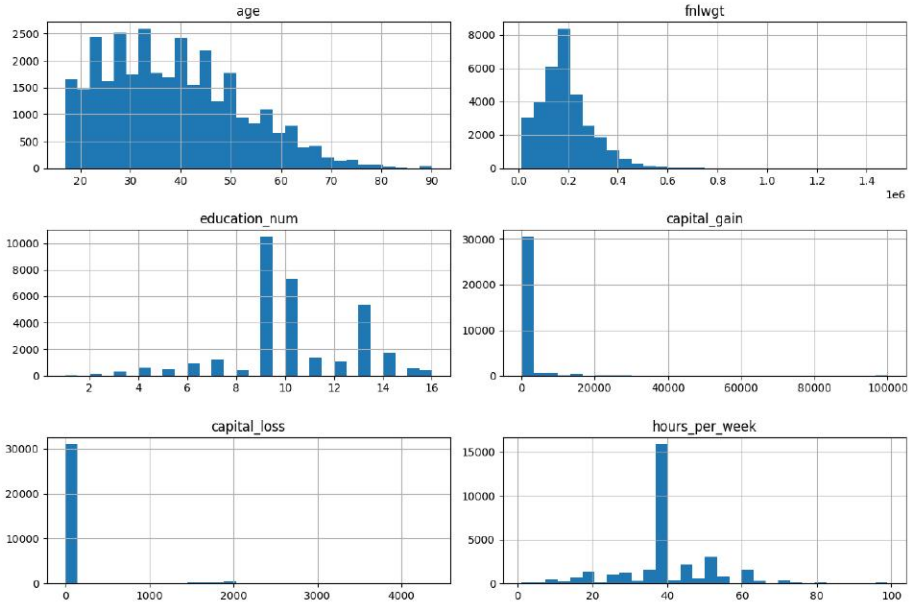

Figure 1: Distributions of numerical variables

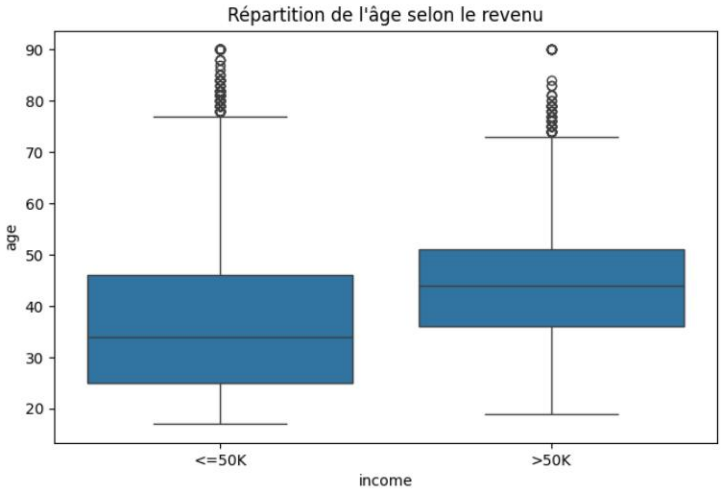

Figure 2: Age distribution by income level

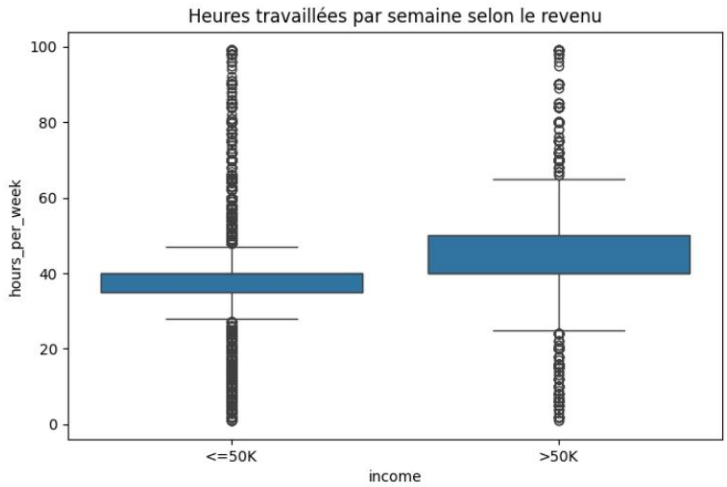

Figure 3: Weekly working hours by income level

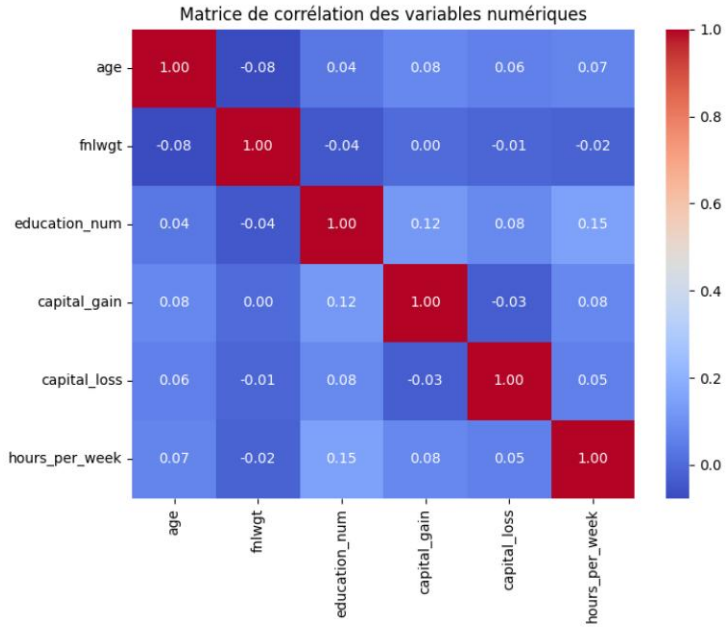

Figure 4: Correlation matrix of numerical variables

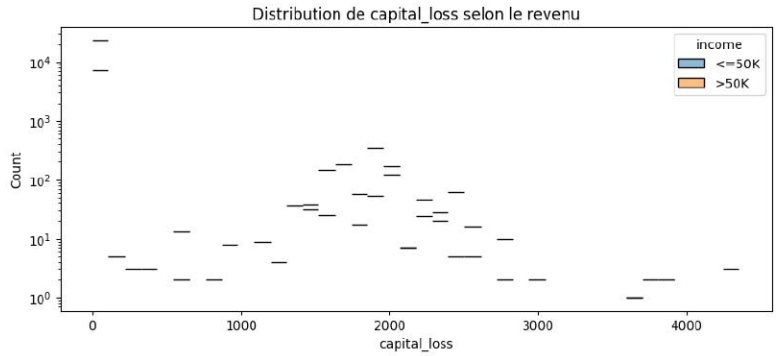

Figure 5: Distribution of capital loss by income level (log scale)

The analysis reveals skewed financial variables, weak linear correlations, and clear differences between income groups, motivating the use of non-linear models.

Phase 1 — Learning curves

Q1.1 — Train / validation gap

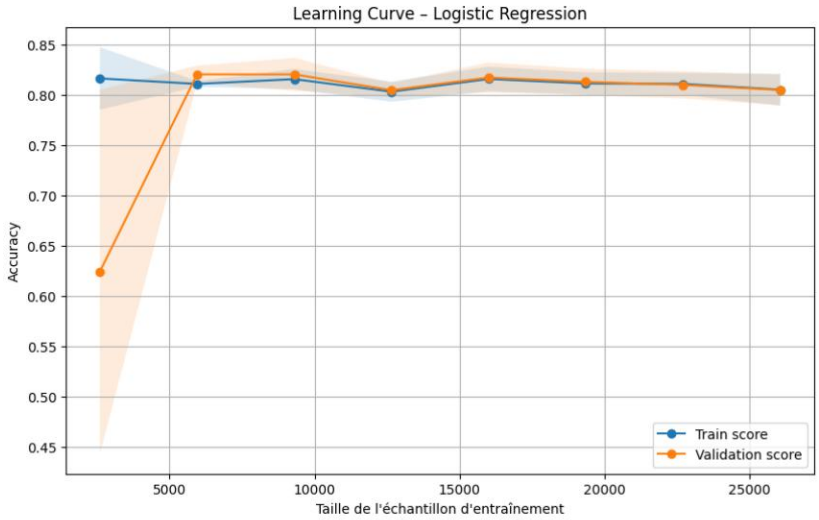

Figure 6: Learning curve — Logistic Regression

The logistic regression exhibits a small train-validation gap, indicating high bias.

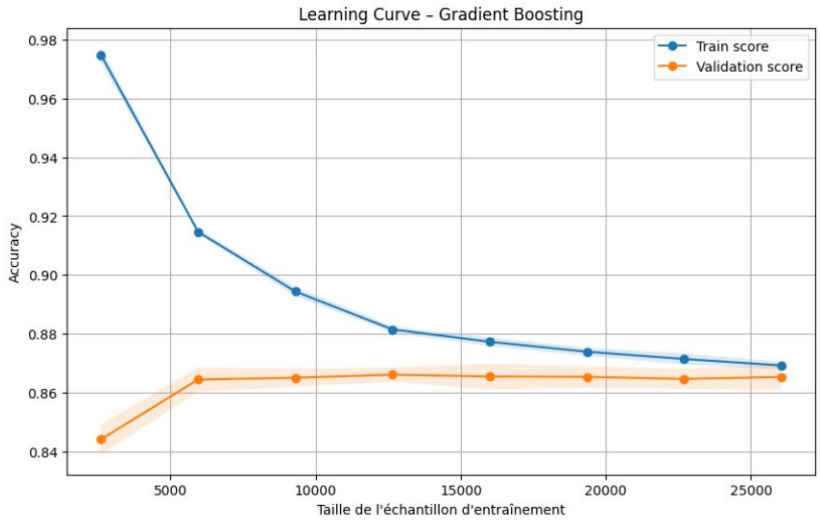

Figure 7: Learning curve — Gradient Boosting

Gradient Boosting shows an initial variance that decreases with more data, leading to better generalization.

Q1.2 — Benefit of additional data

Additional data marginally improve logistic regression but significantly benefit Gradient Boosting, which continues to reduce variance as the training set grows.

Phase 2 — Cross-validation and bootstrap

Q2.1 — Mean performance and variability

Cross-validation accuracy:

- Mean: 0.8654
- Standard deviation: 0.0043

Bootstrap accuracy:

- Mean: 0.8655
- Standard deviation: 0.0029

Q2.2 — Stability analysis

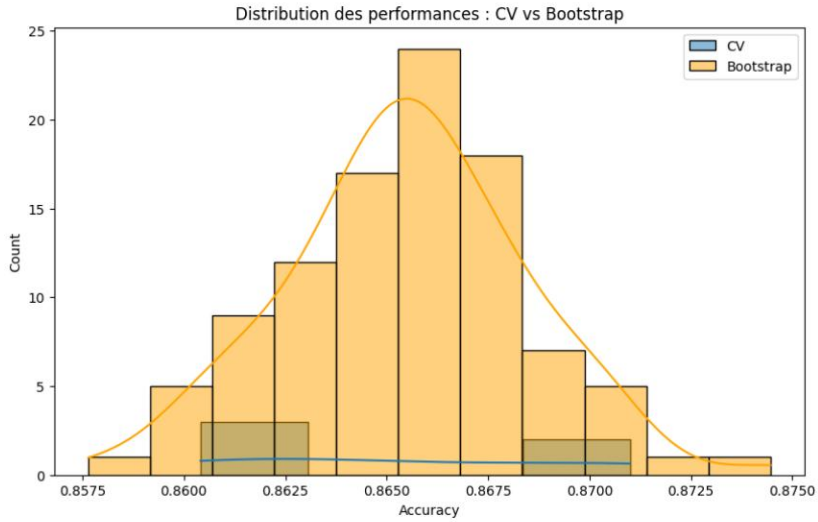

Figure 8: Performance distribution — CV vs Bootstrap

Both methods provide consistent estimates, with bootstrap exhibiting slightly lower variance.

Phase 3 — Hyperparameter tuning

Q3.1 — Search space

The following hyperparameters were optimized: learning rate, number of estimators, maximum depth and subsampling ratio.

Q3.2 — Validation curves

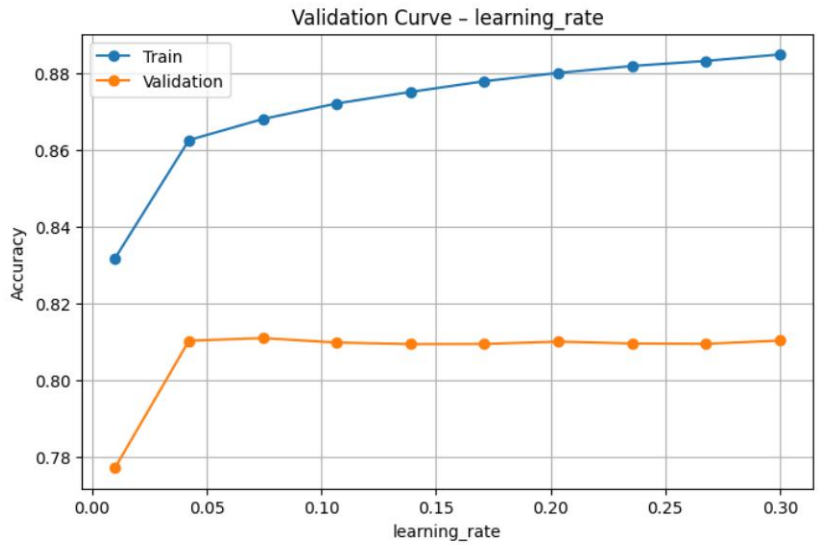

Figure 9: Validation curve — Learning rate

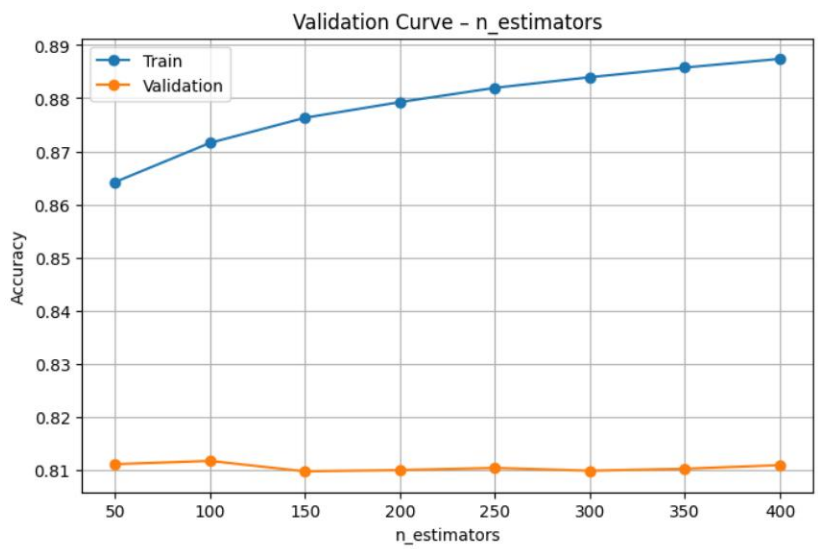

Figure 10: Validation curve — Number of estimators

Q3.3 — Selected model

Best hyperparameters:

- learning rate = 0.065
- max depth = 4
- number of estimators = 267
- subsample = 0.835

The optimized model achieves a mean accuracy of 0.8717 with a standard deviation of 0.0051.

Phase 4 — Model interpretability

Q4.1 — Global feature importance

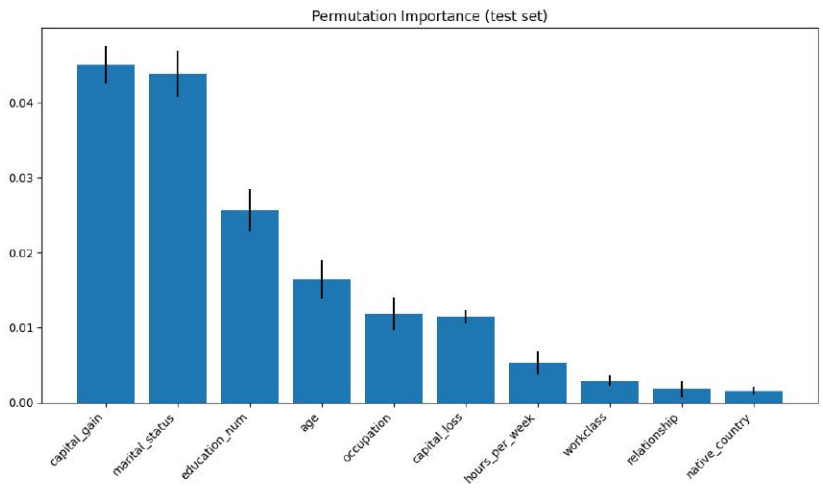

Figure 11: Permutation importance on the test set

Q4.2 — SHAP global explanations

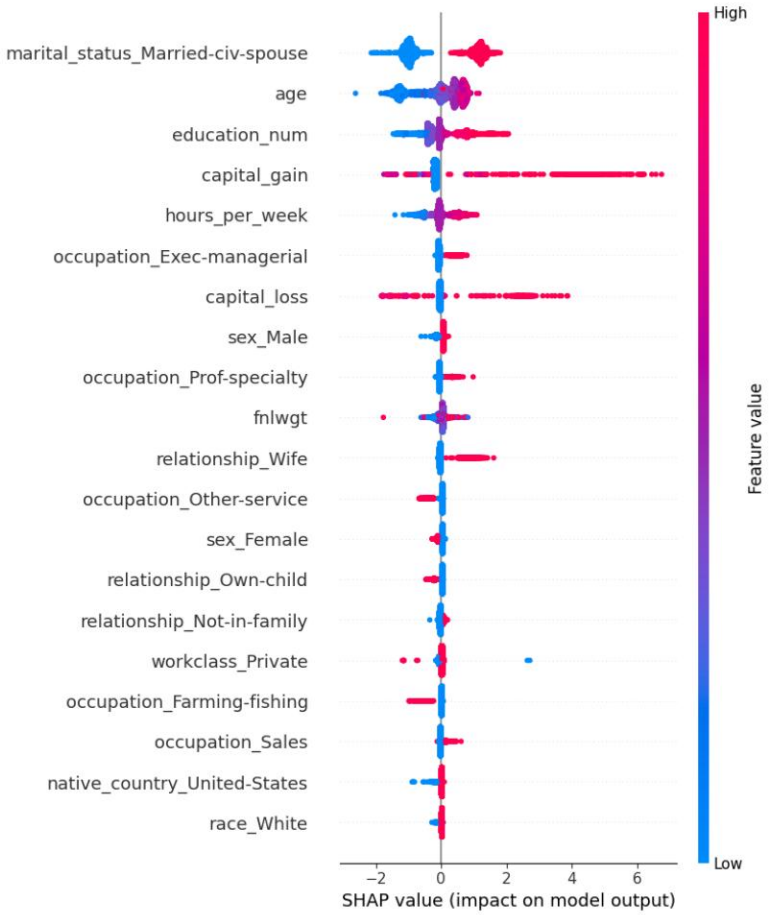

Figure 12: SHAP summary plot

Q4.3 — Feature interactions and local explanations

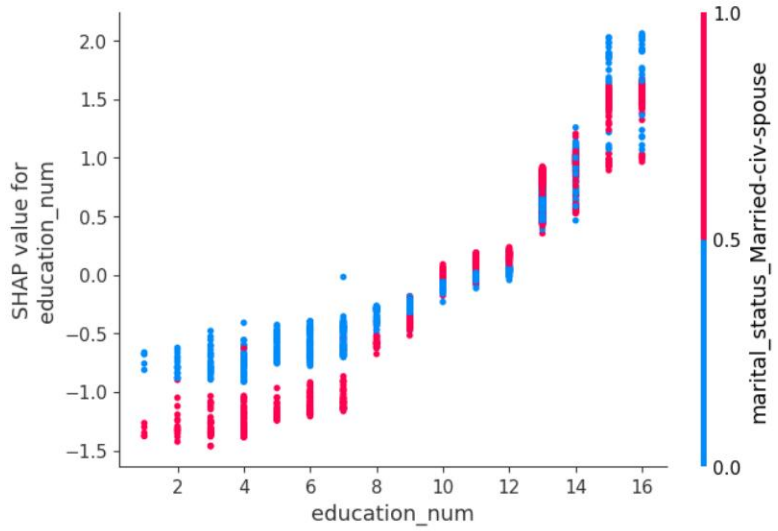

Figure 13: SHAP dependence — education_num

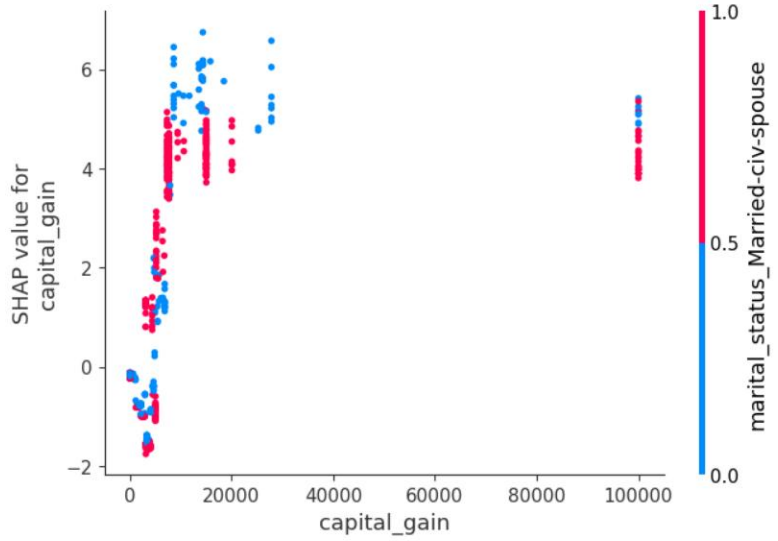

Figure 14: SHAP dependence — capital_gain

Figure 15: Local SHAP explanation

Phase 5 — Data drift analysis

Q5.1 — Drift simulation

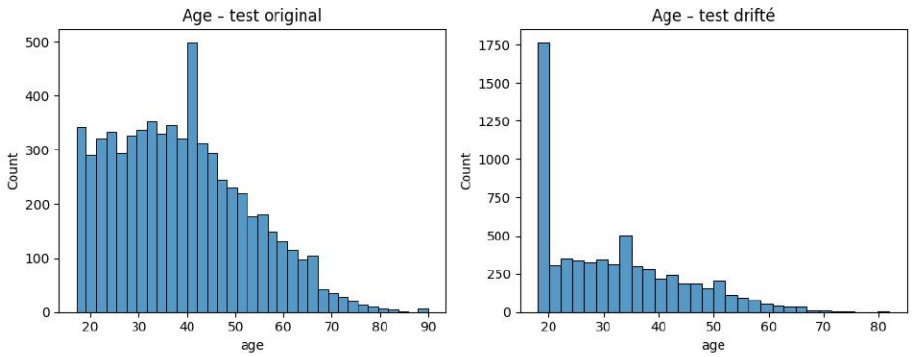

Figure 16: Age distribution — original vs drifted test set

Q5.2 — Drift metrics

Strong drift is detected for `age` and `hours_per_week` using KS, Wasserstein distance and PSI. Categorical drift mainly affects `occupation` and `marital_status`.

Q5.3 — Impact on performance

Under drift, accuracy decreases from 0.867 to 0.827 and balanced accuracy from 0.789 to 0.655, showing strong sensitivity of the minority class.

Phase 6 — Drift mitigation

Q6.1 — Feature suppression

Removing highly drifted variables improves robustness but reduces balanced accuracy to approximately 0.76.

Q6.2 — Retraining

Retraining on the new domain yields better overall performance with balanced accuracy around 0.78.

Q6.3 — Comparison

Retraining provides the best trade-off between robustness and predictive performance.

Phase 7 — Post-mitigation interpretation

Q7.1 — Global importance stability

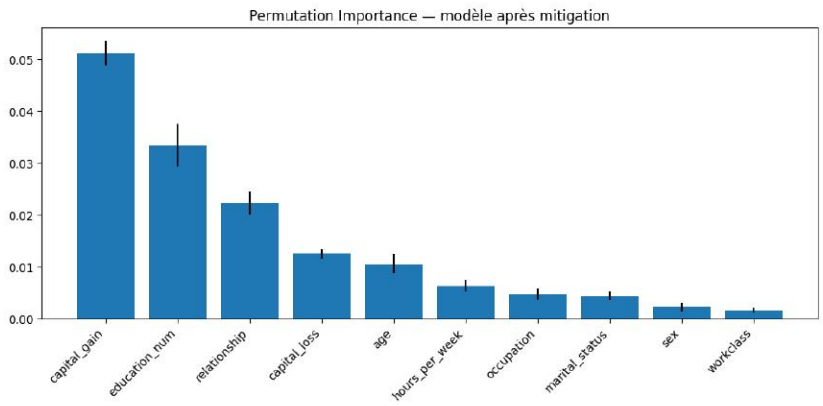

Figure 17: Permutation importance after mitigation

Key features remain dominant, with smoother importance distributions.

Q7.2 — Local explanation stability

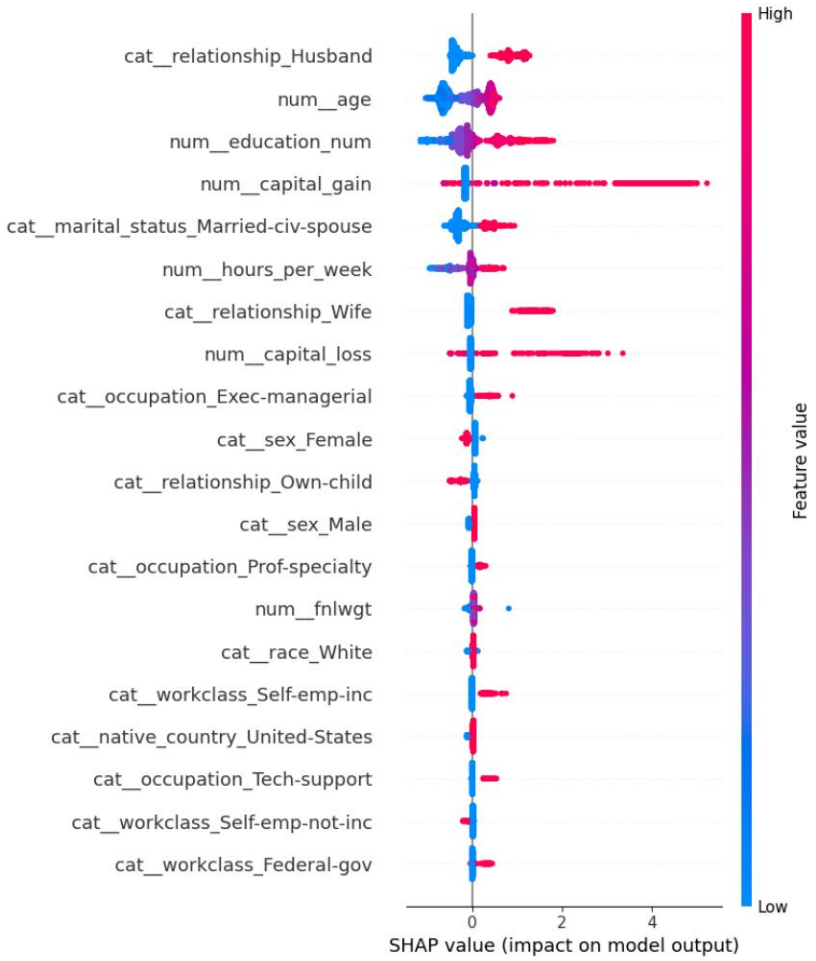

Figure 18: SHAP summary after mitigation

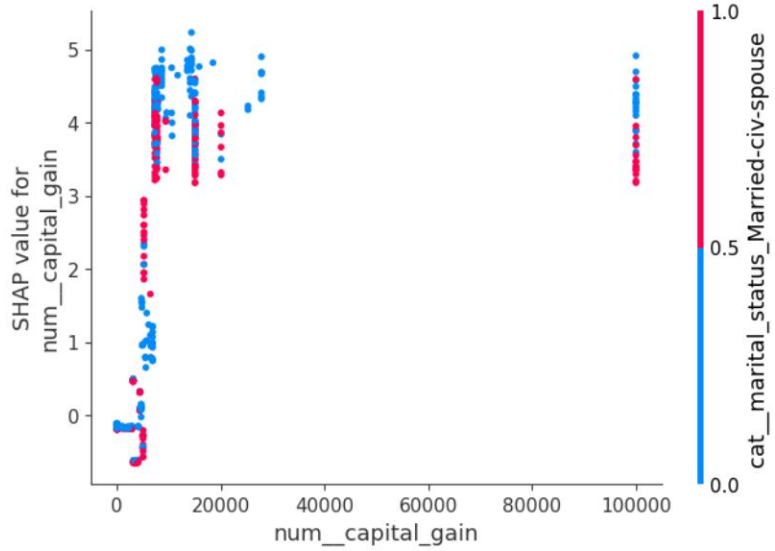

Figure 19: SHAP dependence after mitigation

Figure 20: Local SHAP explanation after mitigation

Local explanations remain coherent, indicating improved robustness without instability.

Phase 8 — Synthesis, monitoring and limitations

Q8.1 — Scenario comparison

Scenario	Accuracy	Balanced Acc	F1
Baseline	0.867	0.789	0.697
Feature removal	0.857	0.760	0.659
Retraining	0.863	0.778	0.684

Q8.2 — Monitoring protocol

Metrics: PSI, KS, chi-squared, balanced accuracy, F1-score. Thresholds: PSI ¿ 0.2 (warning), PSI ¿ 0.5 (critical). Frequency: weekly drift checks, monthly performance evaluation. Actions: analysis, retraining, rollback if severe degradation is detected.

Q8.3 — Limitations

Main limitations include selection bias, undetected complex drift, maintenance cost, and the need for governance in model updates.